Worksheet 1-6 & 1-7 C Algebraic Limits #3

AP Calculus Name_____

Directions: Evaluate the following limits, if possible. Show all work and use proper notation.

1. $\lim_{x\to 1} \frac{x-1}{\sqrt{x}-1}$

2. $\lim_{x\to 3} \sqrt{2x+4}$

3. $\lim_{x\to 0} \frac{x}{\frac{1}{x+3}-\frac{1}{3}}$

4. $\lim_{x\to 1} \frac{\sqrt{x+8}-3}{x-1}$

5. $\lim_{x\to -1} \frac{x+1}{x^2+2x+2}$

6. $\lim_{x\to 0} \frac{\frac{1}{x+2}-\frac{1}{2}}{x}$

7. $\lim_{x\to -1} \frac{x+1}{x^2-2x-3}$

8. $\lim_{x\to 3} \frac{x^2-5x+6}{x-3}$

9. $\lim_{x\to 0} \dfrac{\frac{1}{x-6}+\frac{1}{6}}{x}$

10. $\lim_{x\to 13} \dfrac{\sqrt{x+12}-5}{x-13}$

11. $\lim_{x\to -9^+} \dfrac{x+9}{|x+9|}$

12. $\lim_{x\to 2^-} \dfrac{|x-2|}{3(x-2)}$

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13. Let f be a function where $\lim_{x\to 5} f(x) = \frac{1}{4}$. Which of the following could represent the function f?

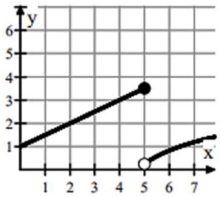
a.)

$$f(x) = \begin{cases} \frac{x-5}{x^2-6x+5} & x \neq 5 \\ 4 & x = 5 \end{cases}$$

b.) The two "pieces" are linear

x	4.8	4.9	4.999	5	5.001	5.1	5.2
$f(x)$	0	0.2	0.249	4	0.251	0.3	0.5

c.)



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14. Let g be a function where $\lim_{x\to 3} g(x) = 6$. Which of the following could represent the function g?

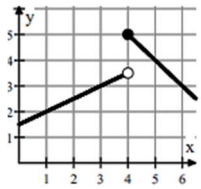
a.)

$$g(x) = \begin{cases} \frac{x^2-x-6}{x-3} & x \neq 3 \\ 6 & x = 3 \end{cases}$$

b.) The two "pieces" are linear

x	2.8	2.9	2.999	3	3.001	3.1	3.2
$f(x)$	6.2	6.01	6.001	6	4.999	4.9	4.8

c.)



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15. If $f(x) = \begin{cases} \frac{(x-3)^2(x^2+1)}{|x-3|} & \text{for } x \neq 3 \\ 4 & \text{for } x = 3 \end{cases}$, then $\lim_{x\to 3} f(x) =$

(A) 0 (B) 2 (C) 10 (D) Nonexistent